

Grunfeld Defense Repertoire

The Exchange center, pressure on d4, development tempi, and dynamic balance



Build a broad center that can survive Black's pressure and learn why every quiet developing move matters.

8 learning units | diagrams | recall tests | engine-checked analysis

ORIENTATION

How to study this volume

PURPOSE

Build a broad center that can survive Black's pressure and learn why every quiet developing move matters.

THE TRAINING LOOP

- First identify Black's defense from pawn placement and move order.
- At every diagram, name the center, the urgent threat, and two candidate moves.
- Hide the recall answer until you have spoken the move and its purpose.
- Replay each line from both sides in Fritz or another board program.
- Review misses after 1, 3, 7, 14, 30, and 60 days.
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WHAT COMPLETE MEANS HERE

This volume contains 8 units inside a connected White repertoire. It covers the main structural and move-order decisions you need for practical play; the PGN companion makes later updating possible.

STRATEGIC FOUNDATION

Principles 1-3

PRINCIPLE 1

Black permits e4 because the center becomes a target.

PRINCIPLE 2

The base on d4 must be defended by pieces, not only pawns.

PRINCIPLE 3

Be3 and Qd2 coordinate development and reduce tactical pressure.

VERIFICATION QUESTION

What would have to be true for one of these principles to be wrong in the current position? Check forcing moves before applying a strategic rule.

STRATEGIC FOUNDATION

Principles 4-6

PRINCIPLE 4

The g7 bishop, ...c5, and the c-file form Black's main attack.

PRINCIPLE 5

A central pawn advance is good only if the squares behind it remain controlled.

PRINCIPLE 6

Simplification can favor White when the remaining center is stable.

VERIFICATION QUESTION

What would have to be true for one of these principles to be wrong in the current position? Check forcing moves before applying a strategic rule.

REPERTOIRE MAP

Branches 1-8

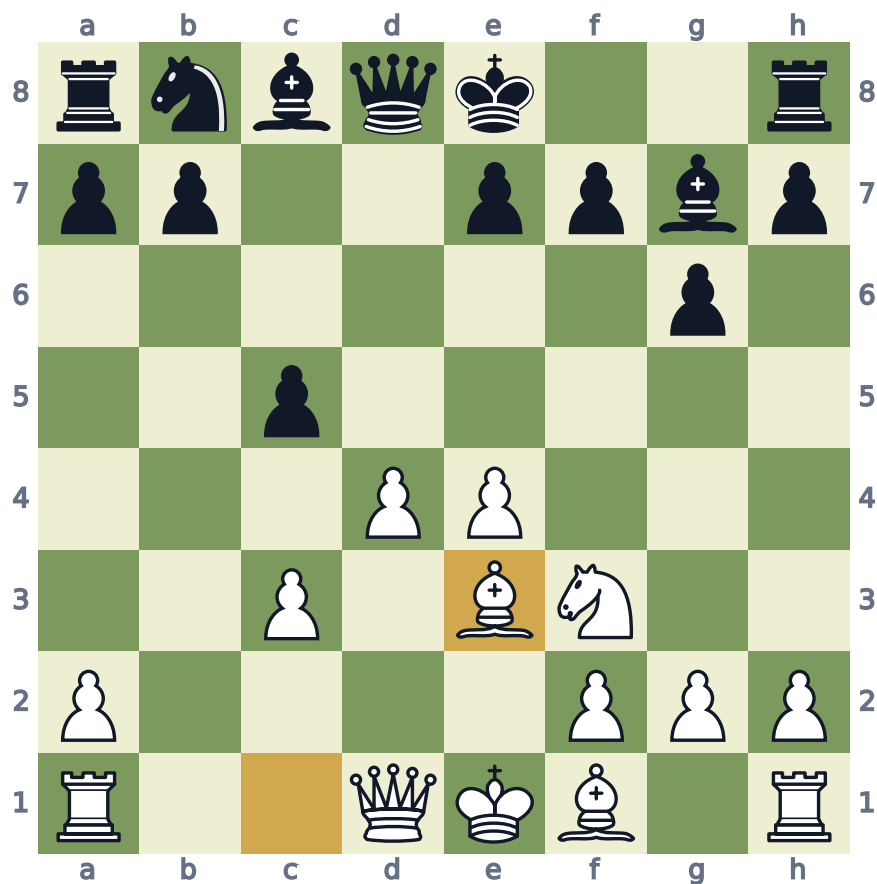
- | | | |
|-----------|--------------------------------|--|
| 01 | Exchange Grunfeld core | After ...Nxc3, the b-pawn strengthens d4 but fixes the structure. |
| 02 | Qd2 development | Qd2 is useful because it serves development and structure at once. |
| 03 | Rb1 system | When the a1 rook is a tactical target, Rb1 can be a developing move. |
| 04 | Early ...c5 | Black's ...c5 is the main proof test of White's center. |
| 05 | Grunfeld ...c6 | Against ...c6, finish development before trying to force e5. |
| 06 | Grunfeld ...Bg4 | Every pin on f3 asks whether d4 will lose a defender. |
| 07 | Grunfeld endgame | An endgame is favorable only if the d4 base and activity survive. |
| 08 | Three-center-pawn setup | f4 is powerful only if White can finish development under pressure. |

NAVIGATION RULE

Identify the pawn structure first, then confirm the move order. A familiar position reached by a different sequence should trigger the same plan.

REPERTOIRE BRANCH 01

Exchange Grunfeld core



COURSE LINE

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Nf3 c5 8. Be3

WHITE'S IDEA

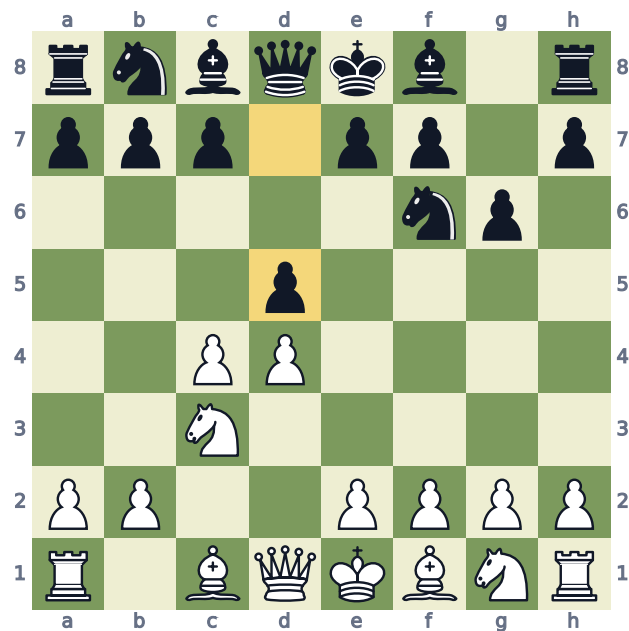
Build the center, develop with tempo, and protect d4.

MAIN DANGER

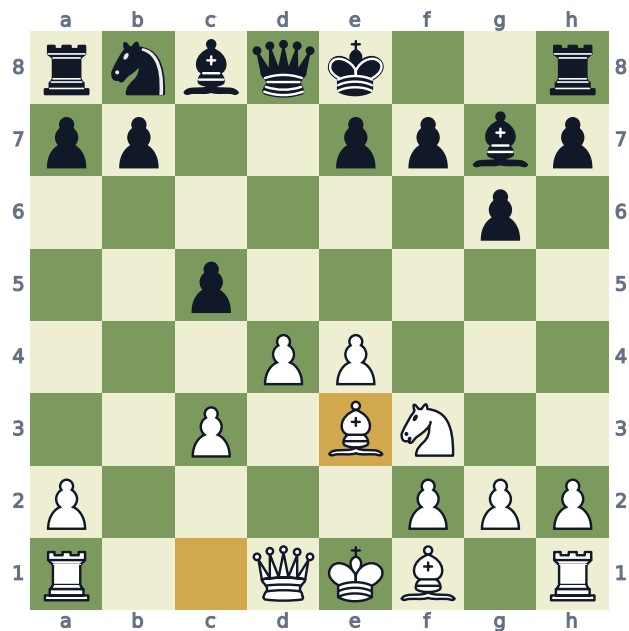
The c3 pawn and d4 base can become targets on the long diagonal.

BRANCH LOGIC 01

Why the moves belong together



AFTER 6 PLIES



COURSE DESTINATION

MEMORY HOOK

After ...Nxc3, the b-pawn strengthens d4 but fixes the structure.

ENGINE CONTINUATION

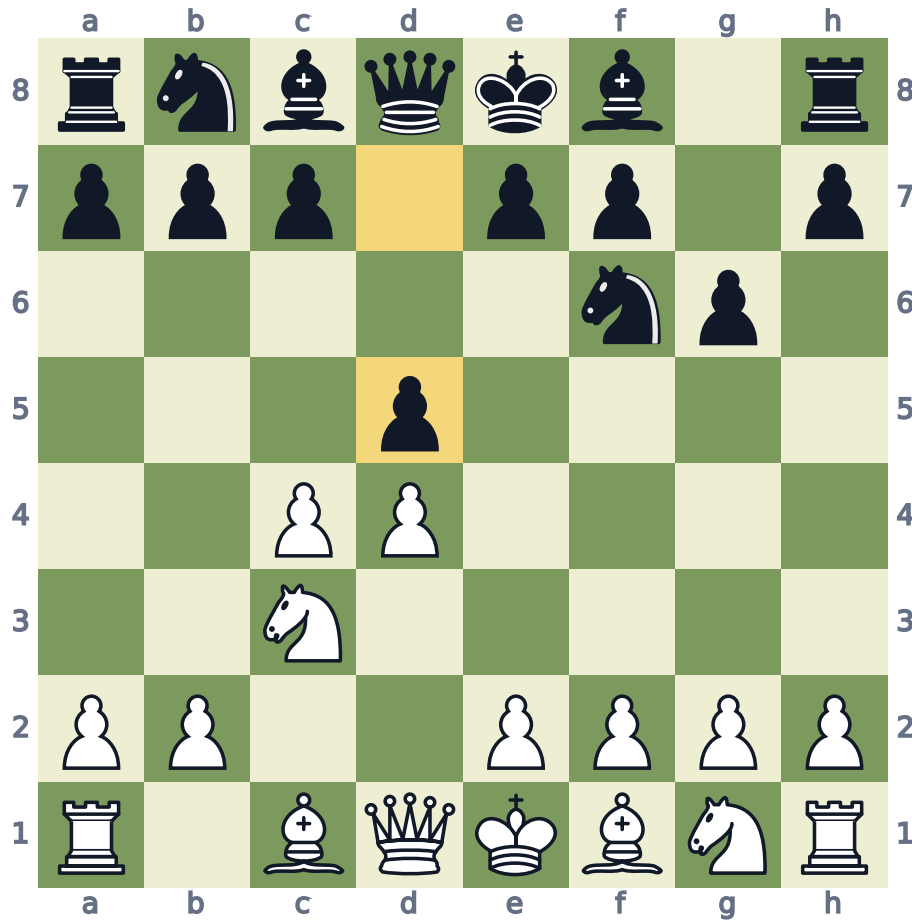
Position estimate: +0.09 for White (equal). Principal continuation: 1. Qa5 Bd2 2. Bg4 Qb3 3. O-O Qxb7 4. Nd7 Qa6 5. Qc7 Rc1

BOUNDARY CONDITION

Use the course line as a reliable destination, but recalculate when Black changes the center, creates a forcing threat, or gains a tactical tempo.

RECALL TEST 01

Exchange Grunfeld core



YOUR TASK

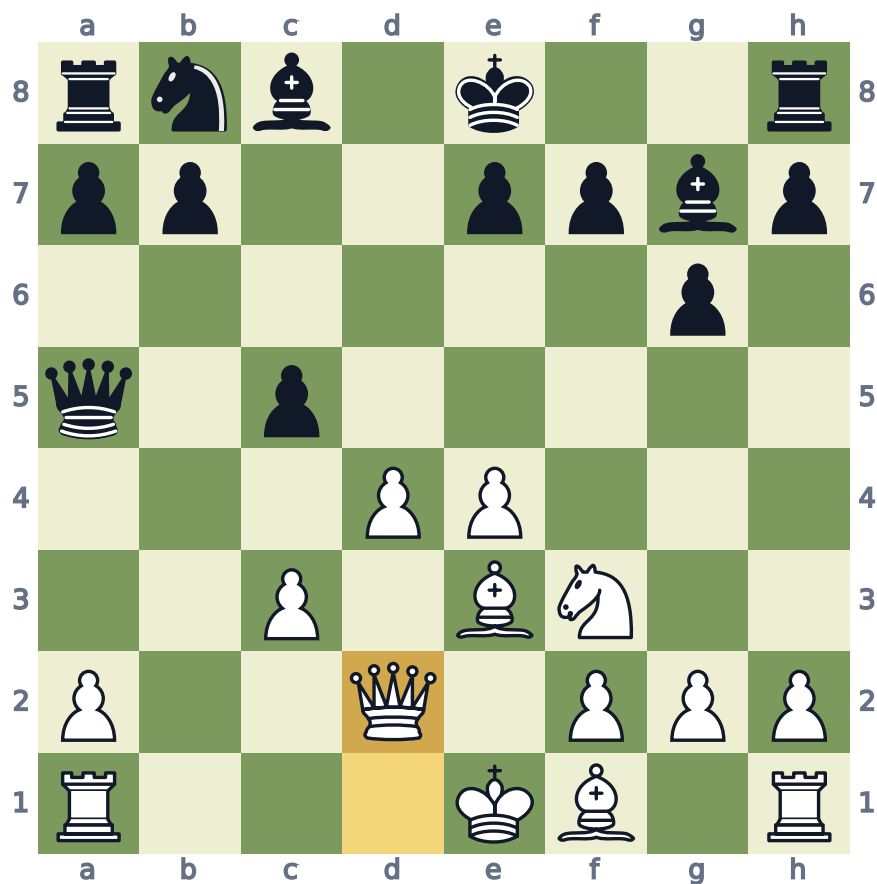
White to move. Name Black's opening setup, choose the repertoire move, and explain its purpose. Then give Black's most challenging reply and one plan for the resulting structure. Do not move the pieces until the full answer is spoken or written.

SCORING

2 move | 2 purpose | 2 Black reply | 2 structure plan | 1 tactical check | 1 notation = 10

REPERTOIRE BRANCH 02

Qd2 development



COURSE LINE

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Nf3 c5 8. Be3 Qa5 9. Qd2

WHITE'S IDEA

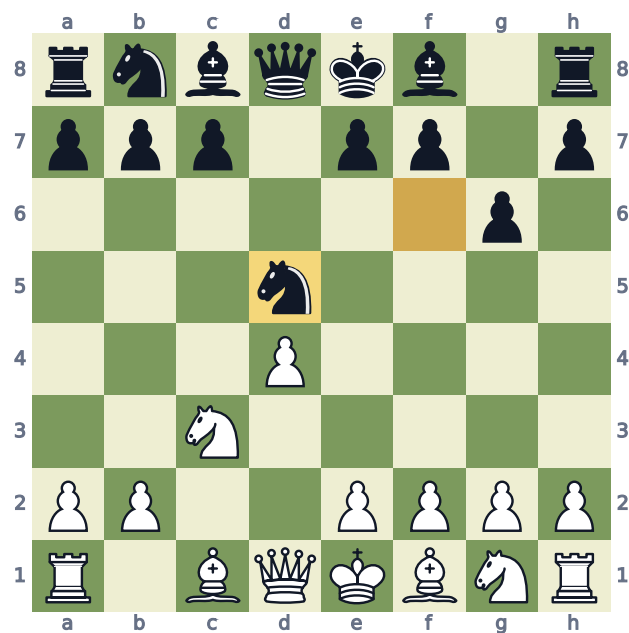
Connect the rooks, defend c3, and prepare Rc1 or Be2.

MAIN DANGER

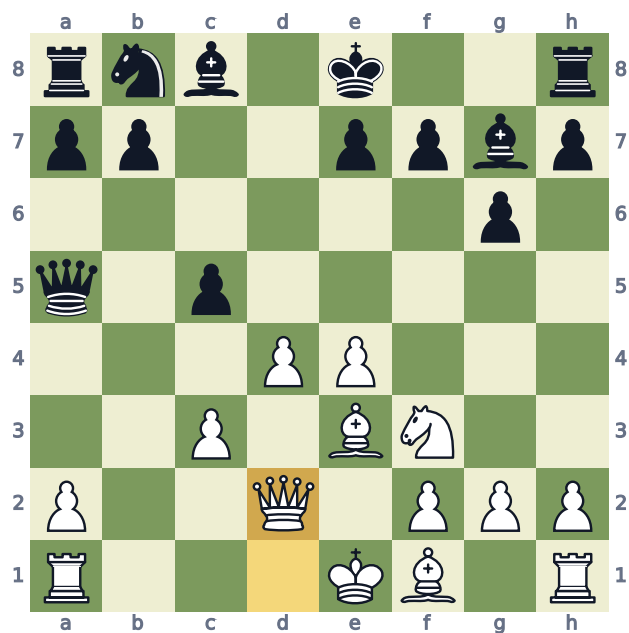
Queen placement can run into ...cxd4 and discovered pressure.

BRANCH LOGIC 02

Why the moves belong together



AFTER 8 PLIES



COURSE DESTINATION

MEMORY HOOK

Qd2 is useful because it serves development and structure at once.

ENGINE CONTINUATION

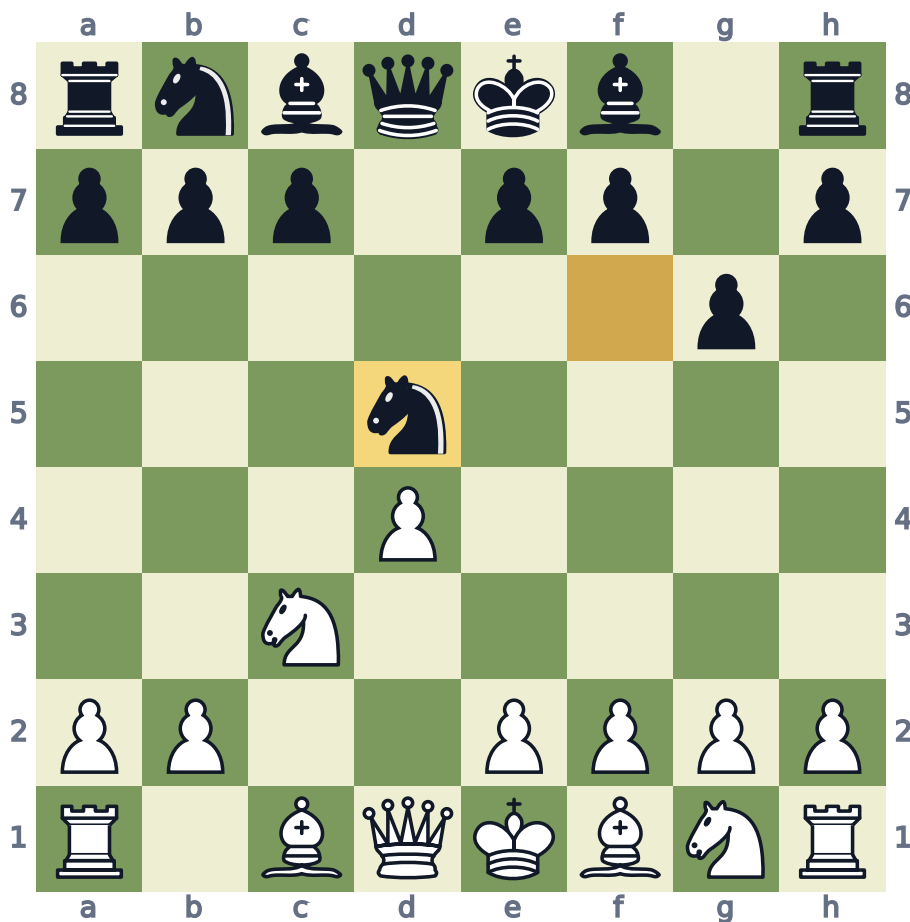
Position estimate: +0.24 for White (White is slightly better). Principal continuation: 1. Nc6 Rb1 2. cxd4 cxd4 3. O-O Qxa5 4. Nxa5 Bd3 5. Rd8

BOUNDARY CONDITION

Use the course line as a reliable destination, but recalculate when Black changes the center, creates a forcing threat, or gains a tactical tempo.

RECALL TEST 02

Qd2 development



YOUR TASK

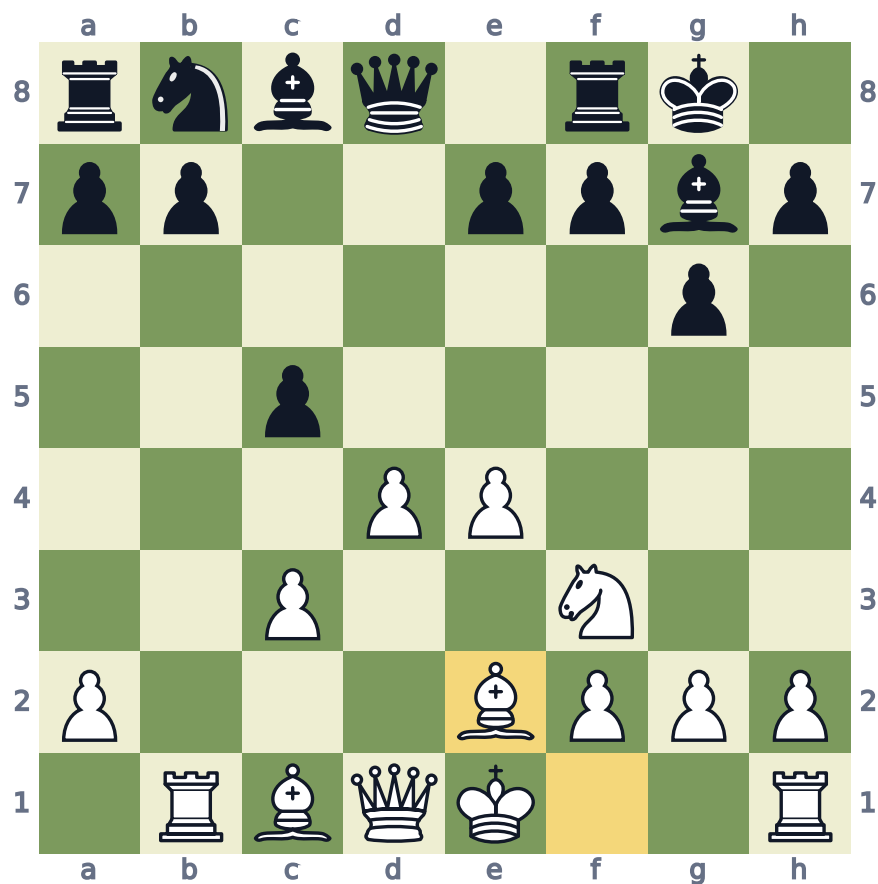
White to move. Name Black's opening setup, choose the repertoire move, and explain its purpose. Then give Black's most challenging reply and one plan for the resulting structure. Do not move the pieces until the full answer is spoken or written.

SCORING

2 move | 2 purpose | 2 Black reply | 2 structure plan | 1 tactical check | 1 notation = 10

REPERTOIRE BRANCH 03

Rb1 system



COURSE LINE

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Rb1 O-O 8. Nf3 c5 9. Be2

WHITE'S IDEA

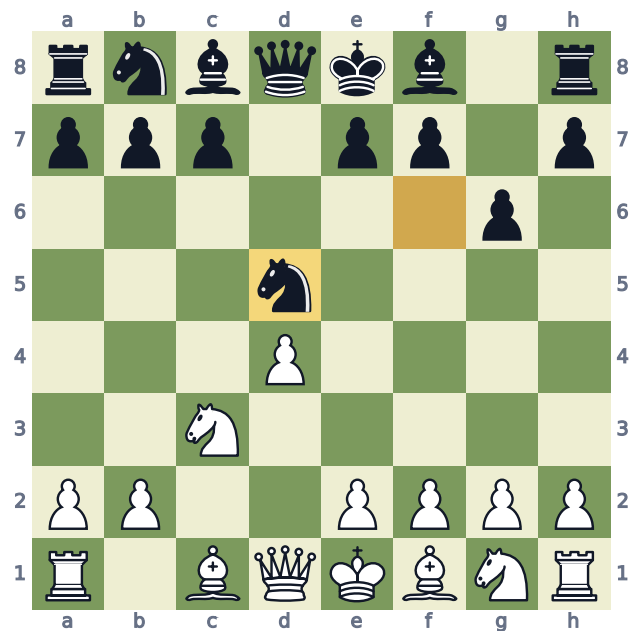
Move the rook off the long diagonal and support queenside expansion.

MAIN DANGER

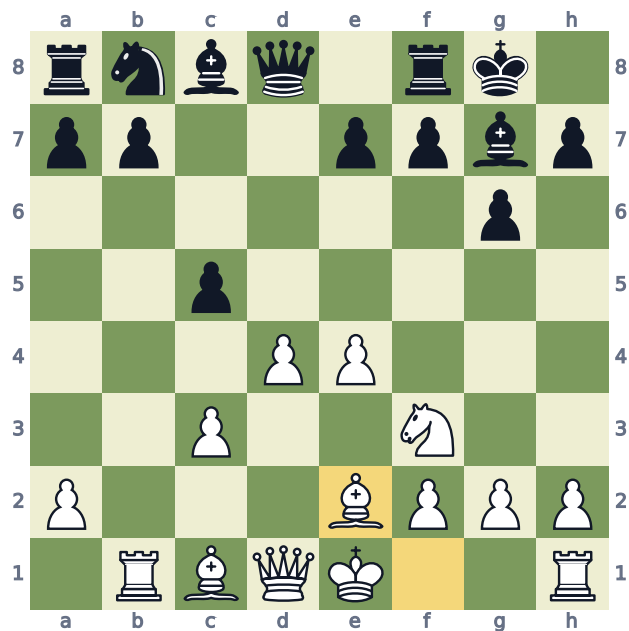
Rb1 costs a tempo if Black can attack the center immediately.

BRANCH LOGIC 03

Why the moves belong together



AFTER 8 PLIES



COURSE DESTINATION

MEMORY HOOK

When the a1 rook is a tactical target, Rb1 can be a developing move.

ENGINE CONTINUATION

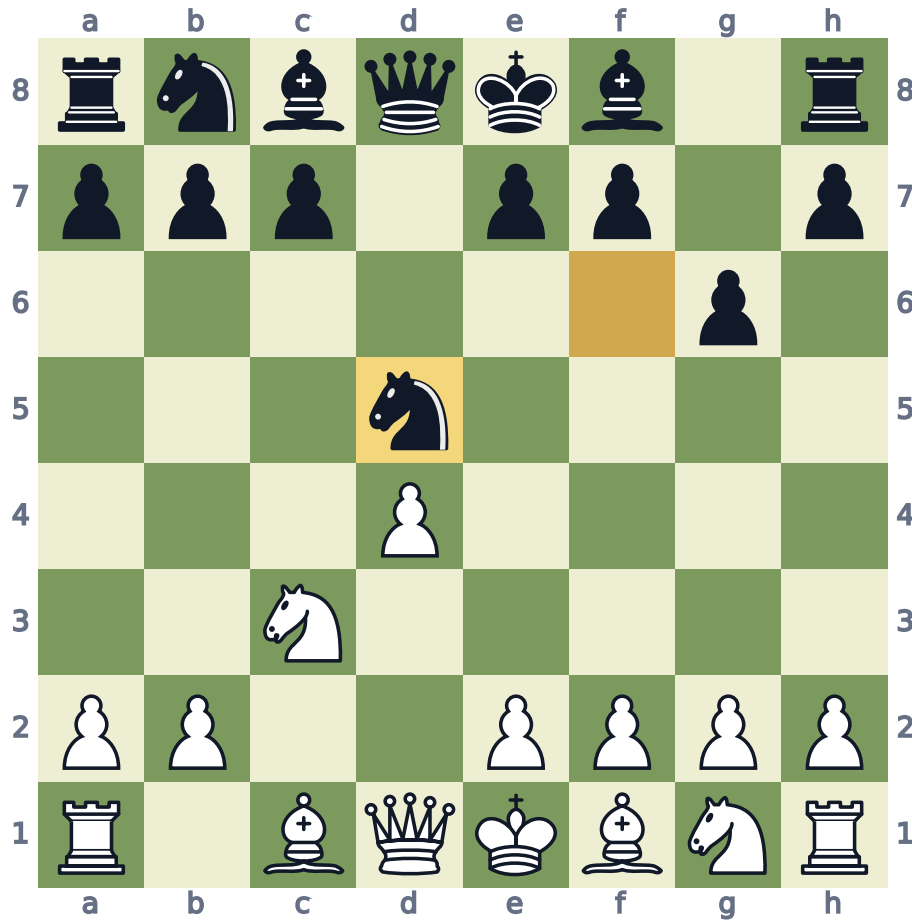
Position estimate: +0.26 for White (White is slightly better). Principal continuation: 1. cxd4 cxd4 2. Qa5+ Bd2 3. Qxa2 O-O 4. Nc6

BOUNDARY CONDITION

Use the course line as a reliable destination, but recalculate when Black changes the center, creates a forcing threat, or gains a tactical tempo.

RECALL TEST 03

Rb1 system



YOUR TASK

White to move. Name Black's opening setup, choose the repertoire move, and explain its purpose. Then give Black's most challenging reply and one plan for the resulting structure. Do not move the pieces until the full answer is spoken or written.

SCORING

2 move | 2 purpose | 2 Black reply | 2 structure plan | 1 tactical check | 1 notation = 10

REPERTOIRE BRANCH 04

Early ...c5



COURSE LINE

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 c5 7. Nf3 Bg7 8. Be3

WHITE'S IDEA

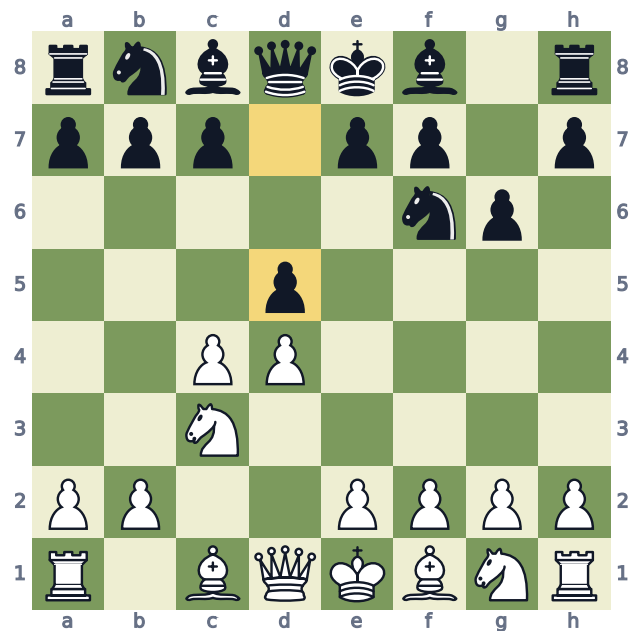
Meet immediate pressure by developing and keeping d4 defended.

MAIN DANGER

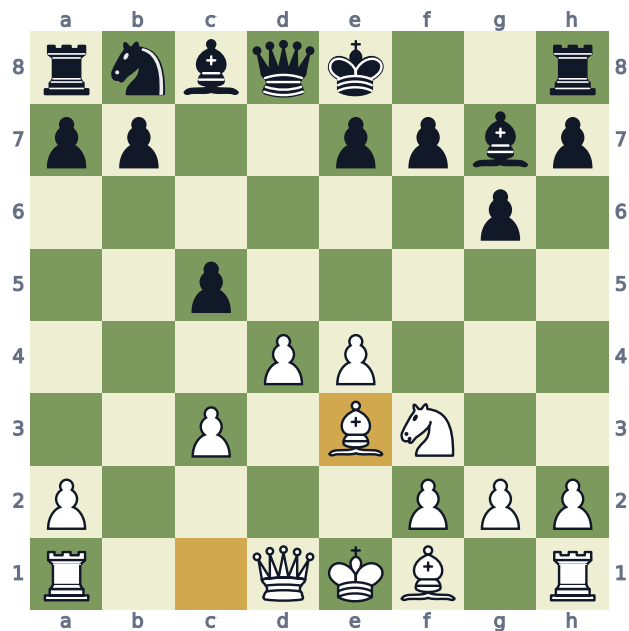
Automatic d5 can surrender key dark squares.

BRANCH LOGIC 04

Why the moves belong together



AFTER 6 PLIES



COURSE DESTINATION

MEMORY HOOK

Black's ...c5 is the main proof test of White's center.

ENGINE CONTINUATION

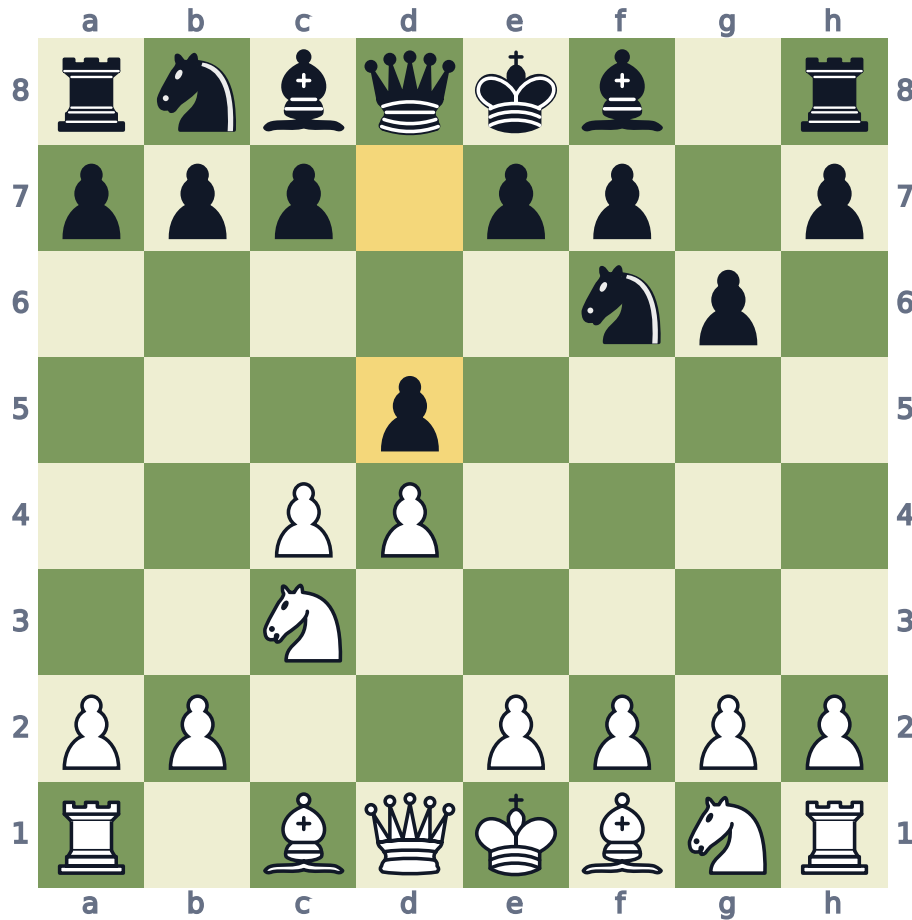
Position estimate: +0.29 for White (White is slightly better). Principal continuation: 1. Qa5 Qd2 2. Nc6 Rb1 3. cxd4 cxd4 4. O-O Bd3 5. Bg4 Qxa5

BOUNDARY CONDITION

Use the course line as a reliable destination, but recalculate when Black changes the center, creates a forcing threat, or gains a tactical tempo.

RECALL TEST 04

Early ...c5



YOUR TASK

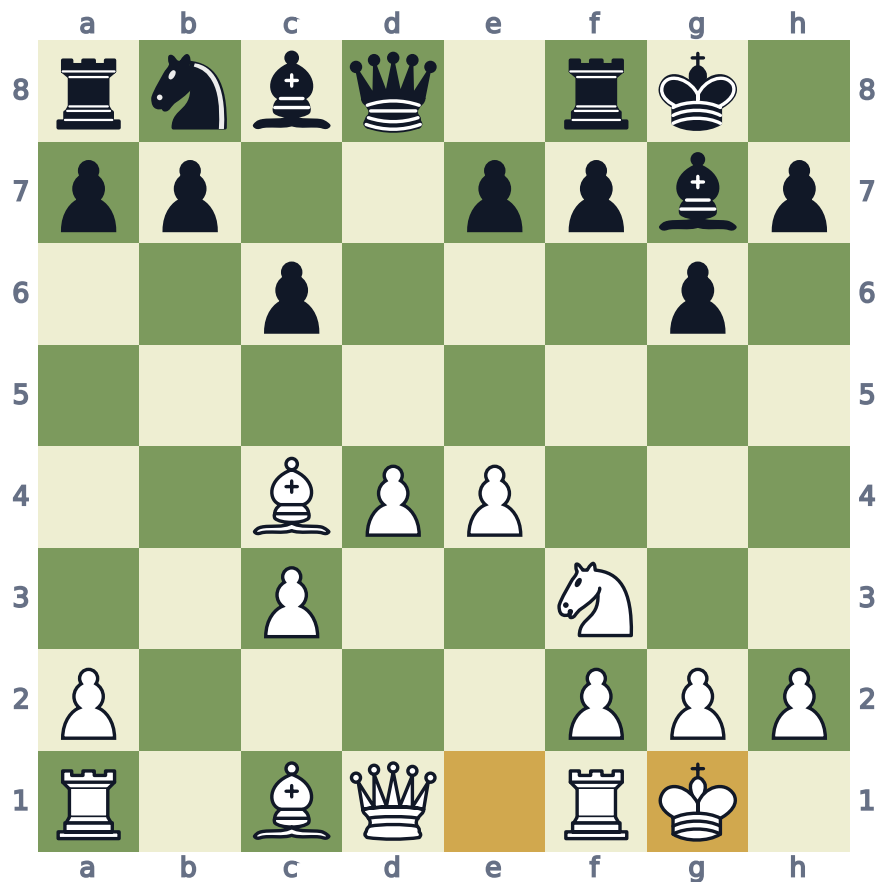
White to move. Name Black's opening setup, choose the repertoire move, and explain its purpose. Then give Black's most challenging reply and one plan for the resulting structure. Do not move the pieces until the full answer is spoken or written.

SCORING

2 move | 2 purpose | 2 Black reply | 2 structure plan | 1 tactical check | 1 notation = 10

REPERTOIRE BRANCH 05

Grunfeld ...c6



COURSE LINE

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Nf3 c6 8. Bc4 O-O 9. O-O

WHITE'S IDEA

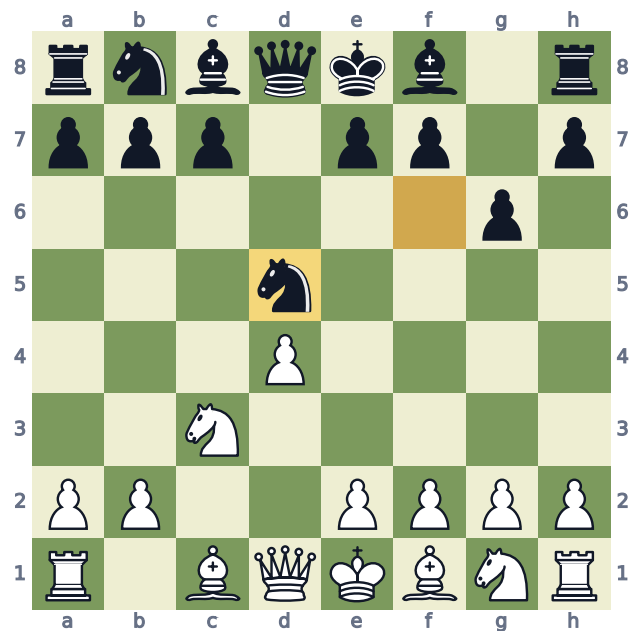
Use the space advantage while Black builds a solid attack on d4.

MAIN DANGER

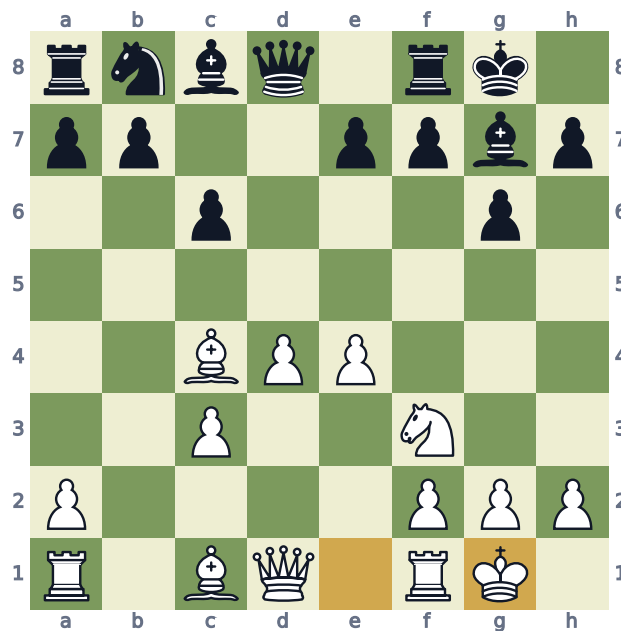
A delayed ...c5 may arrive under better circumstances for Black.

BRANCH LOGIC 05

Why the moves belong together



AFTER 8 PLIES



COURSE DESTINATION

MEMORY HOOK

Against ...c6, finish development before trying to force e5.

ENGINE CONTINUATION

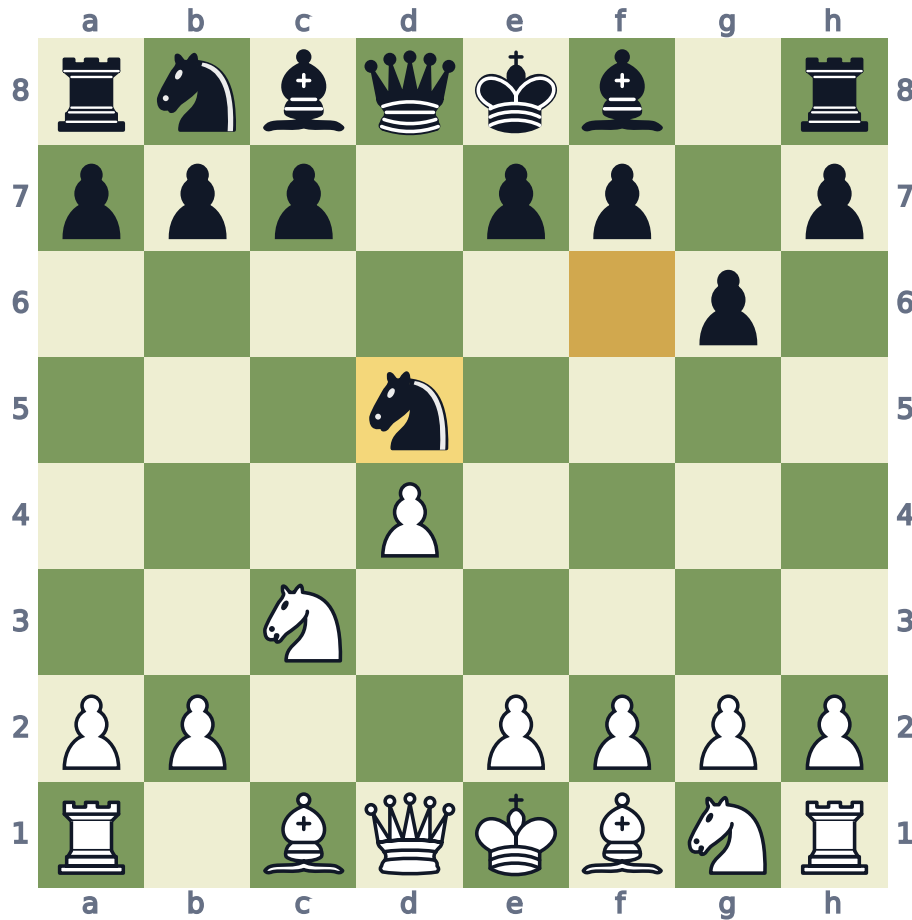
Position estimate: +0.79 for White (White is slightly better). Principal continuation: 1. Bg4 h3 2. Bxf3 Qxf3 3. c5 e5 4. Nc6 Rd1 5. Qc7 Qe4

BOUNDARY CONDITION

Use the course line as a reliable destination, but recalculate when Black changes the center, creates a forcing threat, or gains a tactical tempo.

RECALL TEST 05

Grunfeld ...c6



YOUR TASK

White to move. Name Black's opening setup, choose the repertoire move, and explain its purpose. Then give Black's most challenging reply and one plan for the resulting structure. Do not move the pieces until the full answer is spoken or written.

SCORING

2 move | 2 purpose | 2 Black reply | 2 structure plan | 1 tactical check | 1 notation = 10

REPERTOIRE BRANCH 06

Grunfeld ...Bg4



COURSE LINE

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Nf3 c5 8. Be3 Bg4

WHITE'S IDEA

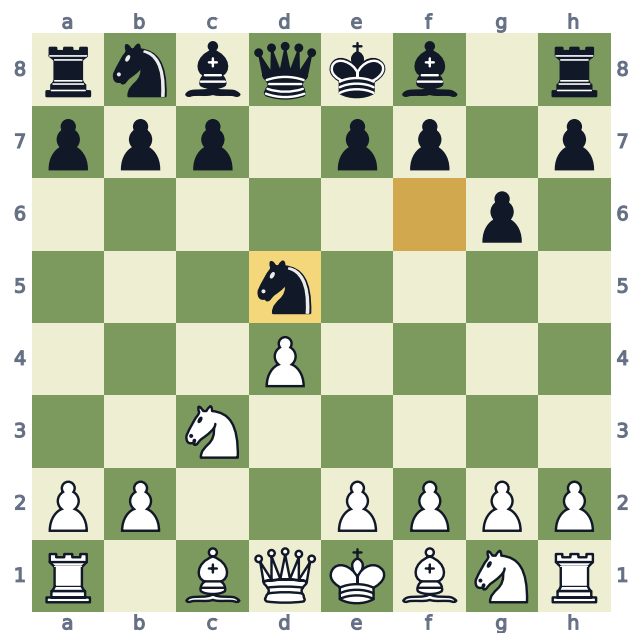
Question the pin with Be2 or Qd2 while preserving d4.

MAIN DANGER

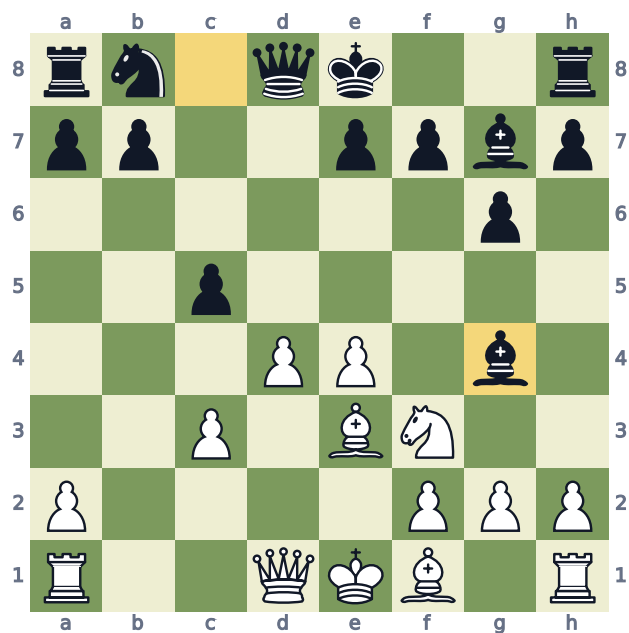
Breaking the pin with f3 can weaken the king and e3.

BRANCH LOGIC 06

Why the moves belong together



AFTER 8 PLIES



COURSE DESTINATION

MEMORY HOOK

Every pin on f3 asks whether d4 will lose a defender.

ENGINE CONTINUATION

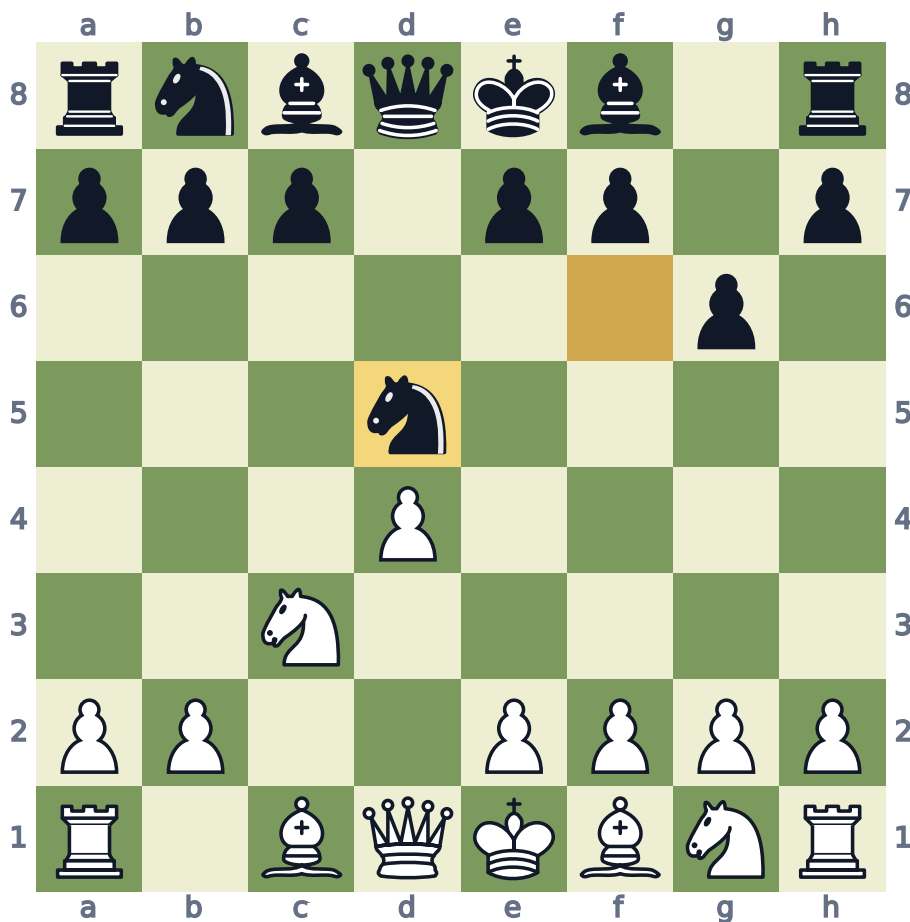
Position estimate: +0.39 for White (White is slightly better). Principal continuation: 1. Rc1 Bxf3 2. gxf3 Qa5 3. Qd2 e6 4. h4

BOUNDARY CONDITION

Use the course line as a reliable destination, but recalculate when Black changes the center, creates a forcing threat, or gains a tactical tempo.

RECALL TEST 06

Grunfeld ...Bg4



YOUR TASK

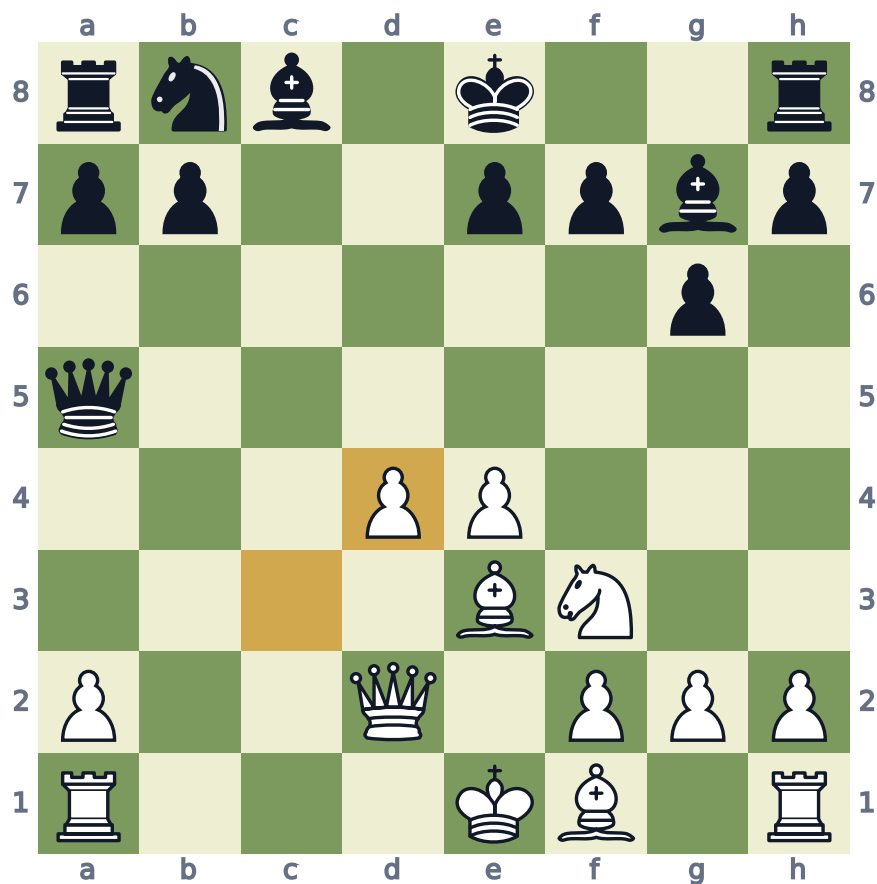
White to move. Name Black's opening setup, choose the repertoire move, and explain its purpose. Then give Black's most challenging reply and one plan for the resulting structure. Do not move the pieces until the full answer is spoken or written.

SCORING

2 move | 2 purpose | 2 Black reply | 2 structure plan | 1 tactical check | 1 notation = 10

REPERTOIRE BRANCH 07

Grunfeld endgame



COURSE LINE

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Nf3 c5 8. Be3 Qa5 9. Qd2 cxd4 10. cxd4

WHITE'S IDEA

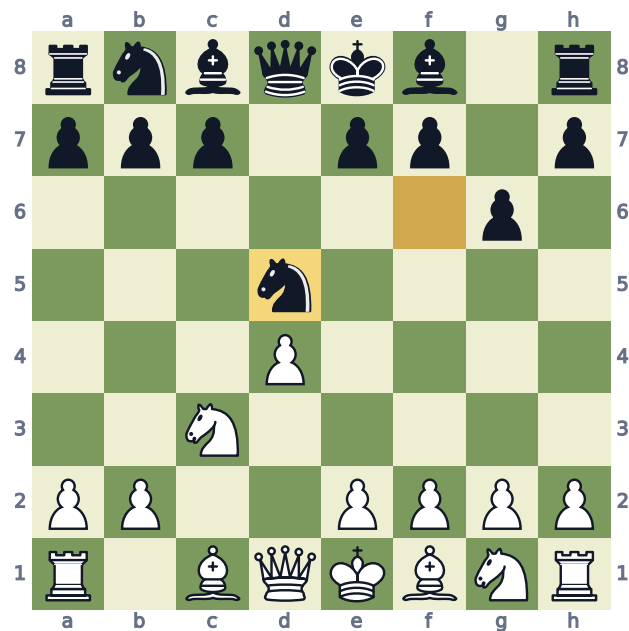
Accept simplification when the center remains supported and space is useful.

MAIN DANGER

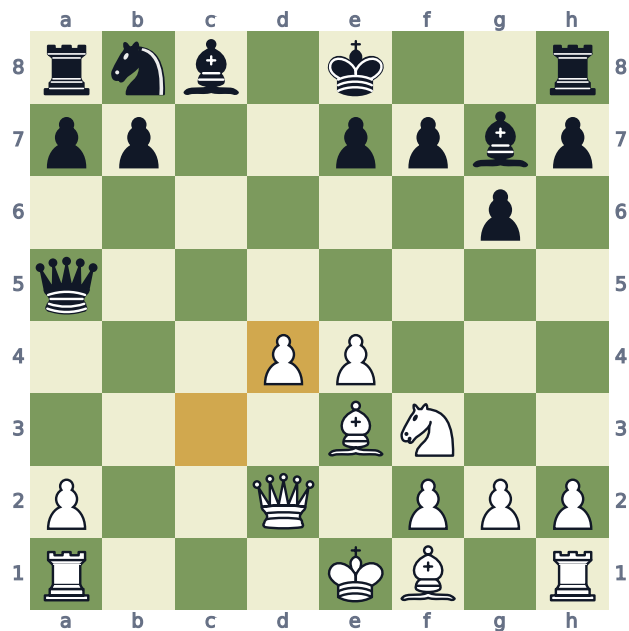
Queenside pawns may become fixed targets after exchanges.

BRANCH LOGIC 07

Why the moves belong together



AFTER 8 PLIES



COURSE DESTINATION

MEMORY HOOK

An endgame is favorable only if the d4 base and activity survive.

ENGINE CONTINUATION

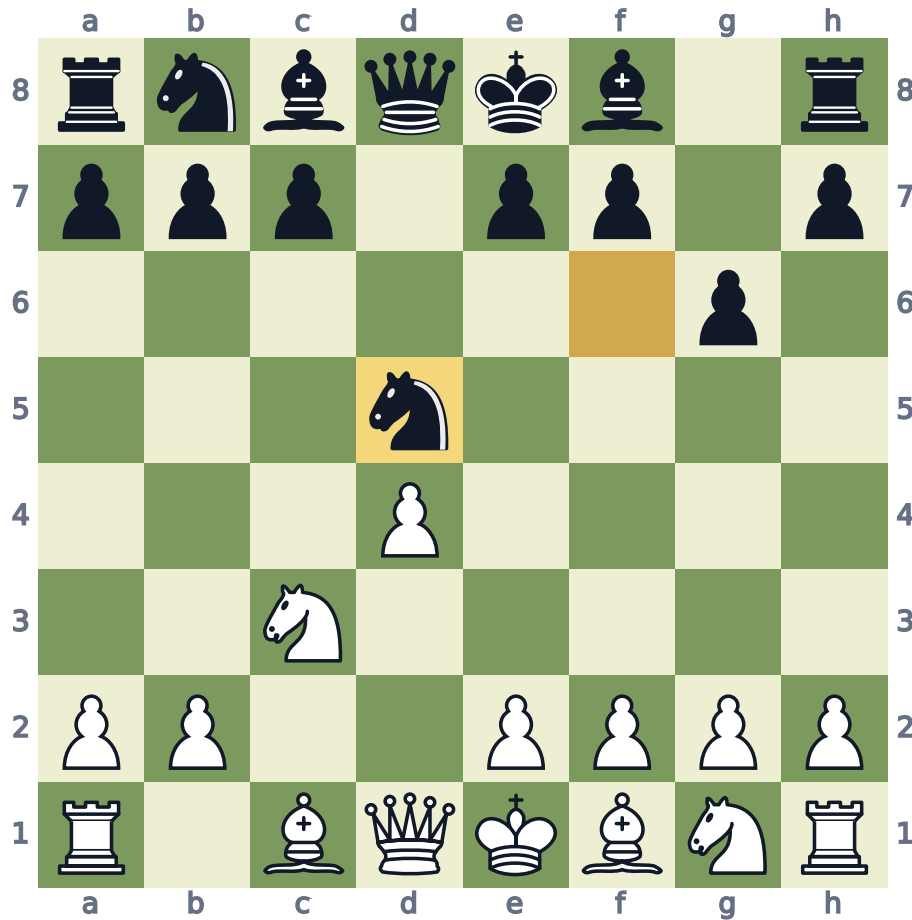
Position estimate: +0.52 for White (White is slightly better). Principal continuation: 1. Qxd2+ Kxd2 2. Nc6 Rb1 3. O-O Bd3 4. Rd8 d5 5. Na5 Ke2

BOUNDARY CONDITION

Use the course line as a reliable destination, but recalculate when Black changes the center, creates a forcing threat, or gains a tactical tempo.

RECALL TEST 07

Grunfeld endgame



YOUR TASK

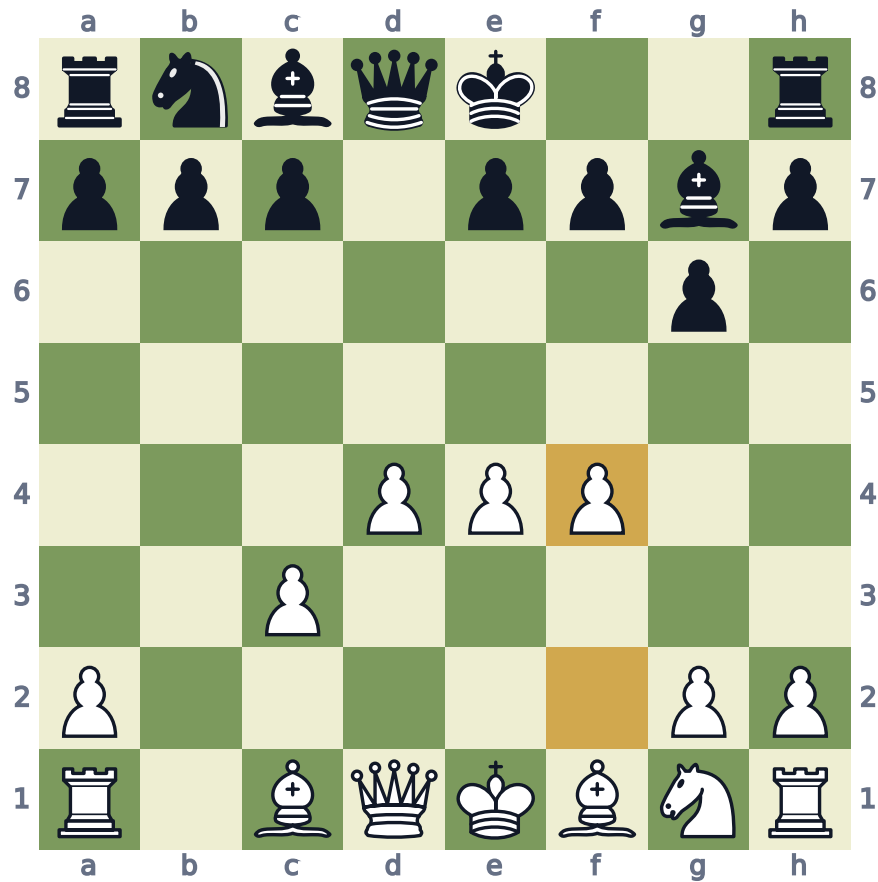
White to move. Name Black's opening setup, choose the repertoire move, and explain its purpose. Then give Black's most challenging reply and one plan for the resulting structure. Do not move the pieces until the full answer is spoken or written.

SCORING

2 move | 2 purpose | 2 Black reply | 2 structure plan | 1 tactical check | 1 notation = 10

REPERTOIRE BRANCH 08

Three-center-pawn setup



COURSE LINE

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. f4

WHITE'S IDEA

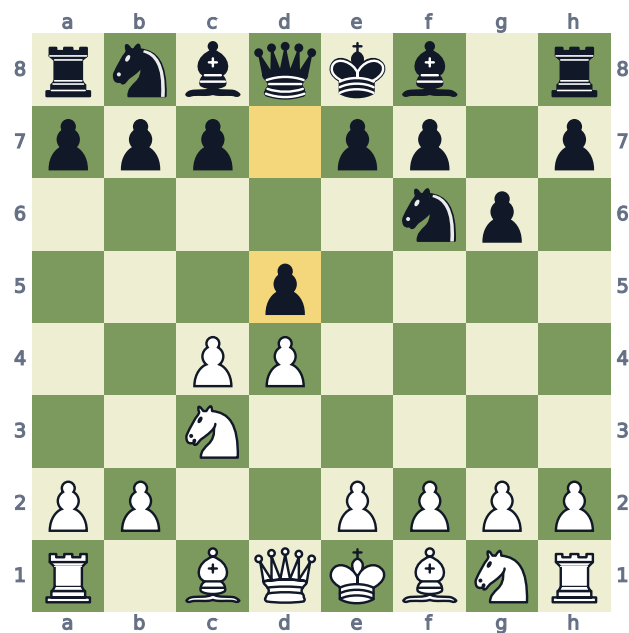
Gain maximal space and prepare e5 while accepting development obligations.

MAIN DANGER

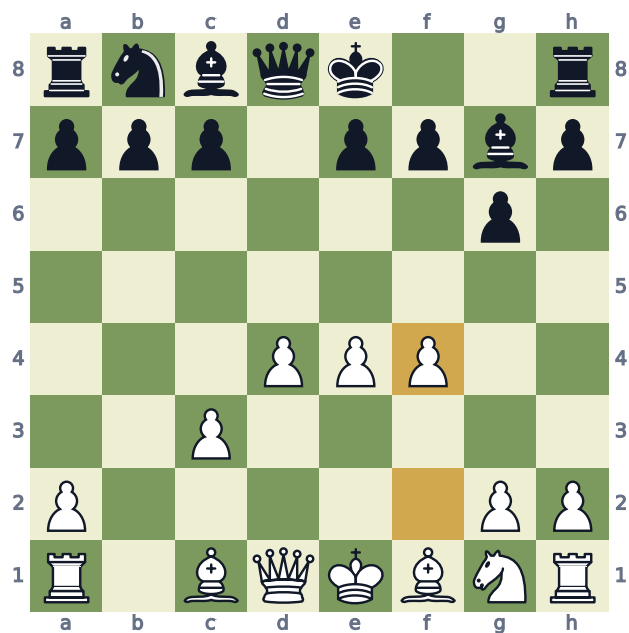
The king and dark squares can become exposed before Nf3.

BRANCH LOGIC 08

Why the moves belong together



AFTER 6 PLIES



COURSE DESTINATION

MEMORY HOOK

f4 is powerful only if White can finish development under pressure.

ENGINE CONTINUATION

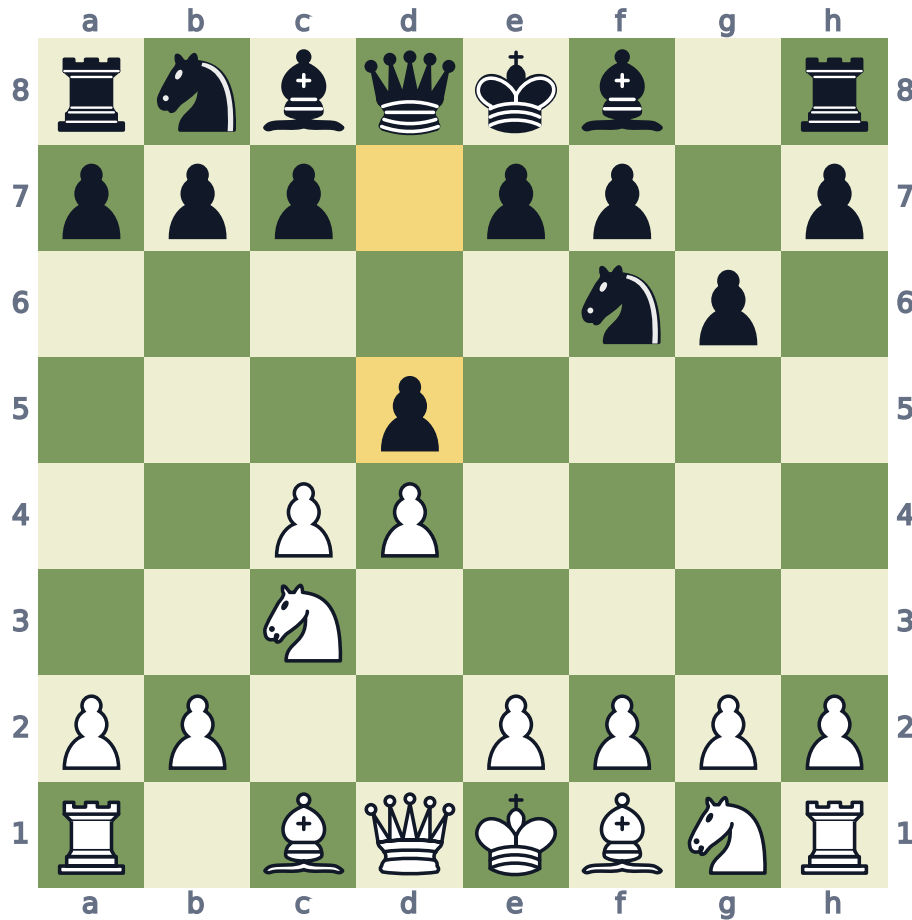
Position estimate: -0.28 for White (Black is slightly better). Principal continuation: 1. c5 Be3 2. Nc6 Nf3

BOUNDARY CONDITION

Use the course line as a reliable destination, but recalculate when Black changes the center, creates a forcing threat, or gains a tactical tempo.

RECALL TEST 08

Three-center-pawn setup



YOUR TASK

White to move. Name Black's opening setup, choose the repertoire move, and explain its purpose. Then give Black's most challenging reply and one plan for the resulting structure. Do not move the pieces until the full answer is spoken or written.

SCORING

2 move | 2 purpose | 2 Black reply | 2 structure plan | 1 tactical check | 1 notation = 10

ANSWER SECTION

Recall solutions begin here

PROTECT THE TRAINING EFFECT

Do not read a solution until you have committed to a move and purpose. A matching move without the correct reason earns partial credit. If the course move is not the engine's first choice, compare plans rather than treating the engine as a memorization oracle.

ERROR LABELS

M = move-order memory | T = tactical oversight | P = plan misunderstanding | V = visualization | C = calculation | R = rushed recognition

REPAIR RULE

Every miss must produce a question you can use in future games. Example: 'When Black supports d5 with c6, which pawn break or recovery method does my repertoire use?'

SOLUTIONS

Answers 1-3

01. EXCHANGE GRUNFELD CORE

Course move: cxd5. Purpose: Build the center, develop with tempo, and protect d4. Engine comparison: 1. Qb3 c6 2. cxd5 cxd5 3. Bf4 Nc6 4. e3 a6 5. Nf3 (+0.48 for White (White is slightly better)). Warning: The c3 pawn and d4 base can become targets on the long diagonal.

02. QD2 DEVELOPMENT

Course move: e4. Purpose: Connect the rooks, defend c3, and prepare Rc1 or Be2. Engine comparison: 1. e4 Nxc3 2. bxc3 Bg7 3. Nf3 c5 4. Bb5+ Nc6 5. d5 a6 (+0.31 for White (White is slightly better)). Warning: Queen placement can run into ...cxd4 and discovered pressure.

03. RB1 SYSTEM

Course move: e4. Purpose: Move the rook off the long diagonal and support queenside expansion. Engine comparison: 1. e4 Nxc3 2. bxc3 Bg7 3. Nf3 c5 4. Bb5+ Nc6 5. d5 a6 (+0.31 for White (White is slightly better)). Warning: Rb1 costs a tempo if Black can attack the center immediately.

SOLUTIONS

Answers 4-6

04. EARLY ...C5

Course move: cxd5. Purpose: Meet immediate pressure by developing and keeping d4 defended. Engine comparison: 1. Qb3 c6 2. cxd5 cxd5 3. Bf4 Nc6 4. e3 a6 5. Nf3 (+0.48 for White (White is slightly better)). Warning: Automatic d5 can surrender key dark squares.

05. GRUNFELD ...C6

Course move: e4. Purpose: Use the space advantage while Black builds a solid attack on d4. Engine comparison: 1. e4 Nxc3 2. bxc3 Bg7 3. Nf3 c5 4. Bb5+ Nc6 5. d5 a6 (+0.31 for White (White is slightly better)). Warning: A delayed ...c5 may arrive under better circumstances for Black.

06. GRUNFELD ...BG4

Course move: e4. Purpose: Question the pin with Be2 or Qd2 while preserving d4. Engine comparison: 1. e4 Nxc3 2. bxc3 Bg7 3. Nf3 c5 4. Bb5+ Nc6 5. d5 a6 (+0.31 for White (White is slightly better)). Warning: Breaking the pin with f3 can weaken the king and e3.

SOLUTIONS

Answers 7-8

07. GRUNFELD ENDGAME

Course move: e4. Purpose: Accept simplification when the center remains supported and space is useful. Engine comparison: 1. e4 Nxc3 2. bxc3 Bg7 3. Nf3 c5 4. Bb5+ Nc6 5. d5 a6 (+0.31 for White (White is slightly better)). Warning: Queenside pawns may become fixed targets after exchanges.

08. THREE-CENTER-PAWN SETUP

Course move: cxd5. Purpose: Gain maximal space and prepare e5 while accepting development obligations. Engine comparison: 1. Qb3 c6 2. cxd5 cxd5 3. Bf4 Nc6 4. e3 a6 5. Nf3 (+0.48 for White (White is slightly better)). Warning: The king and dark squares can become exposed before Nf3.

VOLUME REVIEW

Make the repertoire usable

REVIEW TASK 1

Reconstruct every branch from the first move without looking.

REVIEW TASK 2

For each branch, explain one Black plan and one White plan.

REVIEW TASK 3

Shuffle the recall positions so the chapter title cannot reveal the answer.

REVIEW TASK 4

Play each starting position against Fritz from both sides.

REVIEW TASK 5

Add mistakes from your own games to the PGN as new training positions.

REVIEW TASK 6

Retest the weakest three branches after 1, 3, 7, 14, 30, and 60 days.

GRADUATION STANDARD

Score at least 80 percent twice, one week apart, and explain every move in plain language. Then maintain the volume through your own games instead of rereading it continuously.